Representational Methods on Original Neuroscience Datasets

Experimental Results Report

This report collates four completed experiments using Haxby 2001, the Natural Scenes Dataset, and Pereira 2018. Kriegeskorte 2008 is retained as an original published reference RDM resource. Results are descriptive and should not be treated as population-level inferential claims.

Experiment	Dataset	Primary question
1. Method agreement	Haxby 2001	Do representational estimators recover similar category geometry?
2. Vision model vs brain	NSD + ResNet-18	Which model layer best matches image-level brain responses?
3. Language model vs brain	Pereira + DistilGPT-2	Which transformer layer best matches language responses?
4. Searchlight RSA	Haxby 2001	Where is object-category geometry spatially expressed?

Executive Summary

Haxby category geometry was reproducible across methods and localized strongly across independent run halves. ResNet-18 alignment with the three-session NSD subset was weak and metric-dependent. DistilGPT-2 produced measurable predictive alignment with Pereira responses, but all metrics favored the input embedding rather than a middle layer.

1. Haxby Method Agreement

Input: 12 runs x 8 object categories x 464 ventral-temporal voxels. The experiment compared classic correlation-distance RSA, crossnobis distances, a PCM-derived RDM, and an odd-run reference RDM using Spearman correlations of their upper triangles.

RDM comparison	Spearman rho
Classic RSA vs odd-run reference	0.871
Crossnobis vs PCM	0.765
Classic RSA vs PCM	0.758
Classic RSA vs crossnobis	0.678
PCM vs odd-run reference	0.651
Crossnobis vs odd-run reference	0.517

Run-level results

Mean CKA between runs was 0.393. Mean angular shape distance was 0.707 radians (about 40.5 degrees). Classic RSA generalized most strongly to the independent odd-run reference, while crossnobis and PCM agreed strongly with each other.

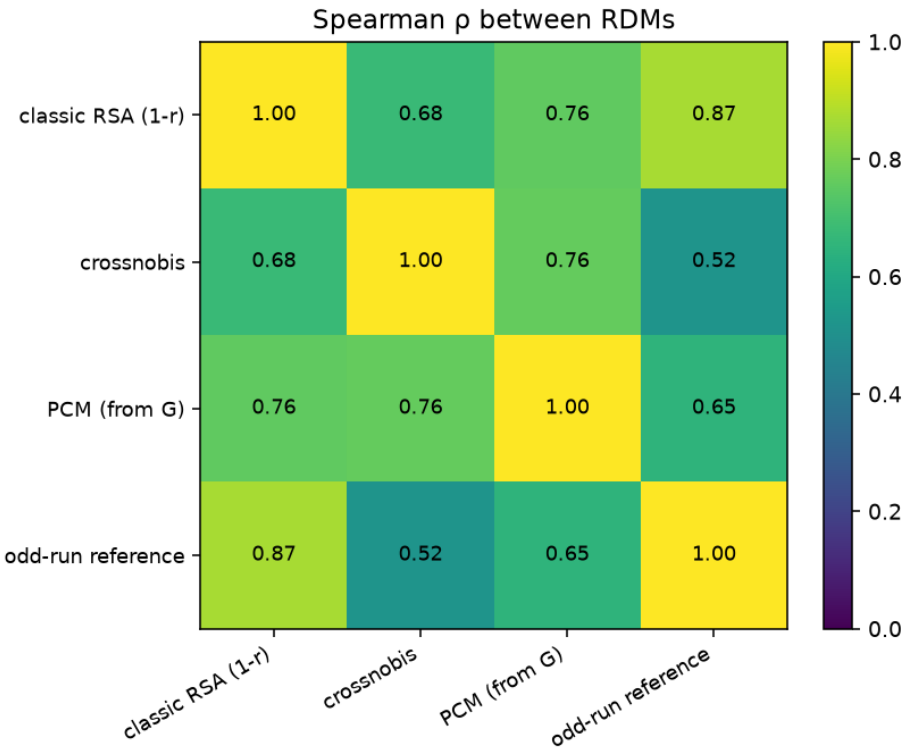


Figure 1. Pairwise RDM agreement on Haxby subject 2.

2. ResNet-18 Versus NSD

Input: 152 exactly matched natural images, three brain repetitions, 512 selected subject-1 voxels, and ImageNet-pretrained ResNet-18 features. Higher is better for RSA, CKA, and ridge correlation; lower is better for shape distance.

Layer	Classic RSA	Crossnobis	CKA	Shape dist.	Ridge r
conv1	0.006	0.001	0.035	1.436	-0.018
layer1	-0.091	-0.101	0.065	1.224	0.000
layer2	-0.054	-0.034	0.074	1.121	0.014
layer3	-0.039	0.002	0.112	1.014	-0.008
layer4	-0.038	-0.030	0.162	0.931	-0.012

Interpretation

Classic RSA favored conv1, crossnobis favored layer3, CKA and shape distance favored layer4, and ridge favored layer2. CKA and reversed shape distance had identical layer rankings, but the absolute effects were weak: RSA was near zero and the best ridge correlation was only 0.014. The normalized plot magnifies these small raw differences.

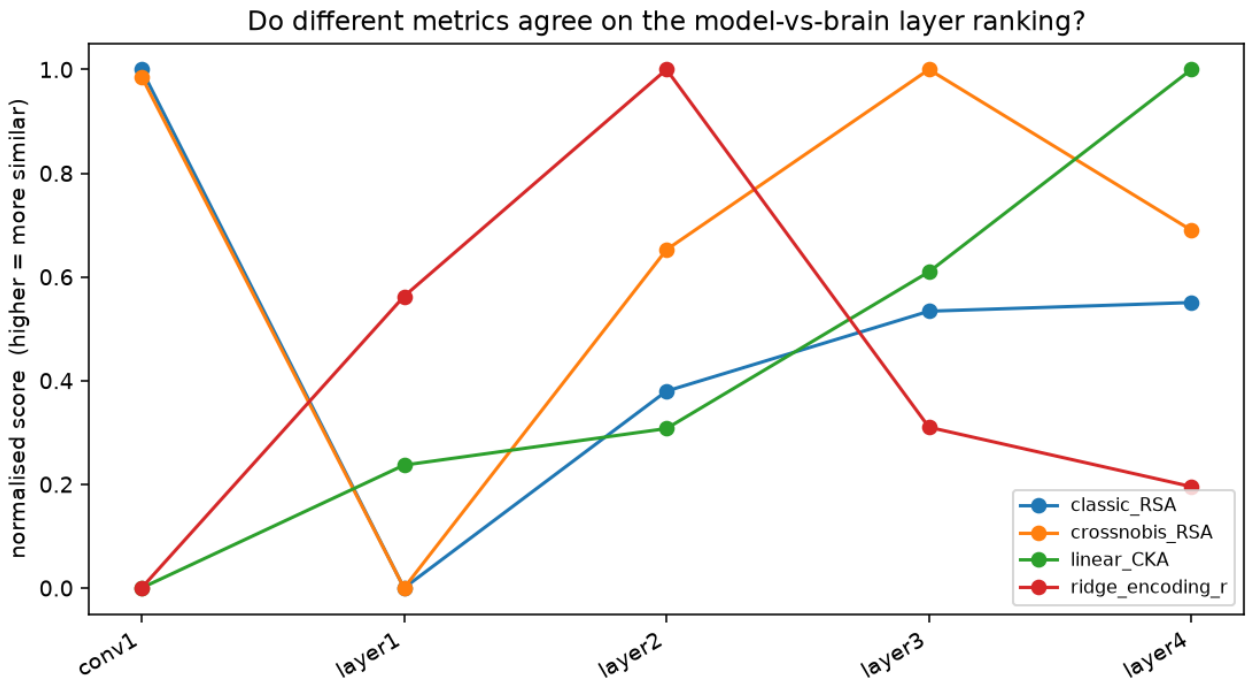


Figure 2. Normalized metric profiles across ResNet-18 layers.

3. DistilGPT-2 Versus Pereira 2018

Input: 384 sentences, 12,155 complete neuroids, and seven DistilGPT-2 representation stages. RSA compares sentence RDMs, CKA compares full representation spaces, and ridge predicts held-out neural responses.

Layer	RSA	CKA	Ridge
embedding	0.042	0.152	0.232
h1	0.035	0.150	0.217
h2	-0.033	0.042	0.202
h3	-0.031	0.045	0.220
h4	-0.030	0.056	0.209
h5	0.009	0.093	0.212
h6	0.034	0.049	0.216

Interpretation

All methods selected the input embedding. RSA and CKA rankings agreed strongly (Spearman rho = 0.893). Ridge scores ranged from 0.202 to 0.232, but RSA and CKA were low. The expected middle-layer peak was not reproduced; architecture, sentence pooling, and multi-subject response construction are plausible contributors.

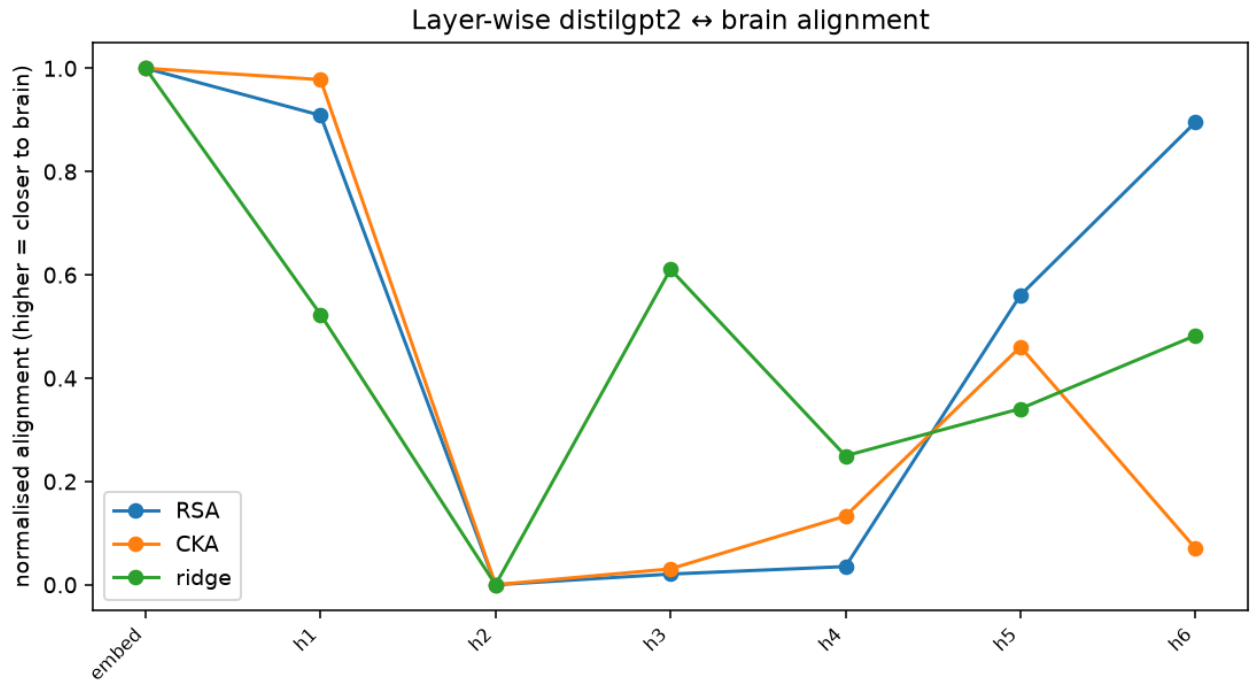


Figure 3. Normalized Pereira alignment across DistilGPT-2 stages.

4. Haxby Searchlight RSA

Odd runs defined an independent category RDM, while even runs supplied local spatial patterns. Each valid ventral-temporal searchlight was compared with the reference using Spearman correlation.

Statistic	Result
Map shape	40 x 64 x 64
Valid centers	462
Positive centers	447 (96.8%)
Median correlation	0.313
95th percentile	0.616
Peak voxel	(30, 21, 29)
Native scanner coordinate	(-36.75, -39.38, -9.38) mm
Peak correlation	0.751

Interpretation

Most valid searchlights positively reproduced the independent reference geometry. The peak coordinate is in native scanner space, not MNI space. The result supports spatially localized and repeatable object-category structure within this subject's ventral-temporal ROI.

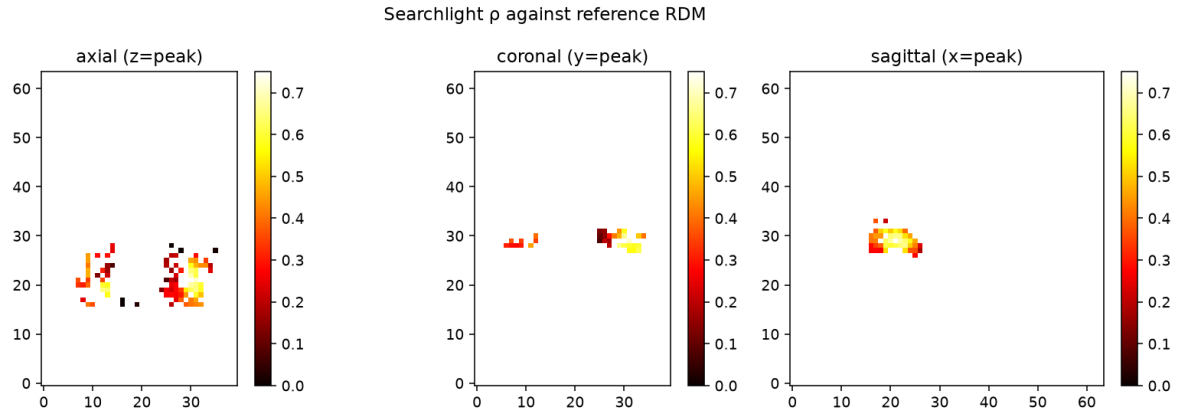


Figure 4. Orthogonal slices through the searchlight peak.

Overall Conclusions

The strongest evidence is the reproducibility and localization of Haxby category geometry. Pereira responses are predictably related to DistilGPT-2 representations, although the input embedding is unexpectedly strongest. The current NSD subset shows little absolute ResNet-brain alignment and substantial metric dependence.

Limitations

No permutation tests, bootstrap confidence intervals, noise ceilings, group-level subject analyses, or corrected searchlight inference were calculated. Ridge results would benefit from nested hyperparameter selection. NSD uses one subject and only three sessions. Haxby is also a single-subject analysis. Accordingly, these results demonstrate the methods and generate hypotheses rather than establish population-level effects.

Recommended Next Steps

Add stimulus-label permutation tests for RSA and CKA; bootstrap stimuli and subjects; estimate split-half noise ceilings; run additional NSD sessions and subjects; use nested cross-validation for ridge; and perform group-level, multiple-comparison-corrected searchlight inference.